Graphs 1 — Gradient · Straight-line graphs · Conversion graphs

You will need squared paper (or a printed grid) for the questions that say *draw*. Marks are shown in brackets. ★ = harder question.

A · GRADIENT

1 Find the gradient of the line joining each pair of points. [4]

a $A(1,\,2)$ and $B(4,\,11)$

b $C(-3,\,4)$ and $D(5,\,-2)$

Gradient $= \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$.

a $\dfrac{11-2}{4-1} = \dfrac{9}{3} = 3$. **b** $\dfrac{-2-4}{5-(-3)} = \dfrac{-6}{8} = -\dfrac{3}{4}$ — negative, the line slopes down from left to right.

2 A wheelchair ramp rises 1.2 m over a horizontal distance of 8 m. [3]

a Work out the gradient of the ramp.

b Building rules say a ramp should have a gradient of at most $\dfrac{1}{12}$. How long (horizontally) must a ramp with the same rise be to meet the rule?

a Gradient $= \dfrac{1.2}{8} = 0.15$ (or $\dfrac{3}{20}$).

b $\dfrac{1.2}{\text{run}} = \dfrac{1}{12} \Rightarrow \text{run} = 1.2 \times 12 = 14.4$ m.

B · STRAIGHT-LINE GRAPHS

3 For the line $y = 3x - 2$: [5]

a Write down its gradient and its y-intercept.

b Copy and complete the table of values for $x = -1,\ 0,\ 1,\ 2,\ 3$.

c Draw the graph for $-1 \le x \le 3$.

d Use your graph (or the equation) to find where the line crosses the x-axis.

a Compare with $y = mx + c$: gradient 3, y-intercept -2, i.e. the point $(0, -2)$.

b $x = -1, 0, 1, 2, 3 \to y = -5, -2, 1, 4, 7$.

c Plot the five points; they lie on a straight line rising 3 for every 1 across.

d On the x-axis $y = 0$: $3x - 2 = 0 \Rightarrow x = \dfrac{2}{3} \approx 0.67$. The crossing point is $\left(\frac{2}{3}, 0\right)$.

4 A line has equation $2x + 5y = 10$. [4]

a Find the points where it crosses the two axes.

b Draw the line.

c Work out its gradient.

> **a** $x = 0 \Rightarrow 5y = 10 \Rightarrow y = 2$, point $(0, 2)$. $y = 0 \Rightarrow 2x = 10 \Rightarrow x = 5$, point $(5, 0)$.
> **b** Join $(0, 2)$ to $(5, 0)$ and extend.
> **c** From $(0, 2)$ to $(5, 0)$: $\dfrac{0 - 2}{5 - 0} = -\dfrac{2}{5}$. (Rearranging: $y = -\frac{2}{5}x + 2$ gives the same.)

5 Here are four lines: $y = 2x + 1$, $2y = x + 6$, $y - 2x = 5$, $x + 2y = 4$. [3]
Which two lines are parallel? Explain how you know.

> Write each as $y = mx + c$: $y = 2x + 1$ ($m = 2$); $y = \frac{1}{2}x + 3$ ($m = \frac{1}{2}$); $y = 2x + 5$ ($m = 2$); $y = -\frac{1}{2}x + 2$ ($m = -\frac{1}{2}$). Parallel lines have equal gradients, so $y = 2x + 1$ and $y - 2x = 5$ are parallel.

C · REAL-LIFE AND CONVERSION GRAPHS

6 A taxi charges a fixed €3 plus €1.50 per kilometre, so the cost C (euros) of a journey of d km is $C = 3 + 1.5d$. [6]

a Draw the graph of C against d for $0 \le d \le 20$.

b What is the gradient of the graph, and what does it represent?

c Use your graph to find how far you can travel for €21.

> **a** Three points are enough for a straight line: $(0, 3)$, $(10, 18)$, $(20, 33)$. Put d on the horizontal axis and C on the vertical axis.
> **b** Gradient 1.5 — the cost per kilometre (€/km). The C-intercept 3 is the fixed charge.
> **c** Read across from $C = 21$: $d = 12$ km. Check: $3 + 1.5 \times 12 = 21$ ✓

7 5 miles is approximately 8 kilometres. [5]

a Draw a conversion graph from miles to kilometres for 0 to 50 miles.

b Use it to convert 35 miles to kilometres.

c Use it to convert 100 km to miles.

> **a** A straight line through the origin and $(50,\ 80)$ (since 50 miles $= 80$ km). Miles on the horizontal axis.
> **b** 35 miles $\to 56$ km (exactly $35 \times \frac{8}{5}$).
> **c** 100 km $\to 62.5$ miles ($100 \times \frac{5}{8}$). Readings from a graph are approximate — 62 or 63 miles is fine.

8 A candle is 24 cm tall and burns down at a steady 2 cm per hour. [5]

a Write an equation for the height h cm of the candle t hours after it is lit.

b Draw the graph of h against t for $0 \leq t \leq 12$.

c Use your graph to find when the candle is a quarter of its original height.

> **a** $h = 24 - 2t$.
> **b** Straight line from $(0, 24)$ down to $(12, 0)$ — gradient -2.
> **c** A quarter of 24 cm is 6 cm: $24 - 2t = 6 \Rightarrow t = 9$ hours.

Total: 35 marks

Check: gradient = rise ÷ run, with the sign · three well-spaced points for a straight line · first-named variable on the vertical axis · label axes with units.